ENTERPRISE DATABASE CLOUD BACKUP & MAINTENANCE

Customer Server Installation, Operations, Monitoring & Uninstallation Guide · v5.0.0

Software Version	Target Audience	Client Runtime	Encryption Standard
v5.0.0 Production	Server Admins / DBA / IT Staff	Zero Python Required	AES-256-GCM + RSA-4096

> **[SECURITY GUARANTEE]** ZERO PYTHON REQUIRED & ZERO CLOUD CREDENTIALS: This software runs natively on Windows Server using built-in Windows PowerShell 5.1+ and .NET Cryptography. No Python runtime or compilers are needed. The client server holds NO Google passwords, OAuth secrets, or service account keys.

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1. Overview & The Three Autonomous Maintenance Modules

This software suite automates three essential database maintenance and protection operations on your Windows Server:

Maintenance Module	Engine Script	Operational Capabilities
Module 1: Database Backup Engine	backup_agent.ps1	Creates native compressed SQL dumps (.bak), encrypts client-side using AES-256-GCM + RSA-4096 into .dbk2 archives, and uploads directly to your dedicated cloud folder.
Module 2: Server Cleanup Scan	storage_monitor.ps1	Monitors disk capacity, enforces retention policies by deleting staging files older than RETENTION_DAYS, and initiates emergency cleanup if disk exceeds threshold (85%).
Module 3: Database Performance & Re-indexing	performance_query.ps1	Samples table index fragmentation, executes a pre-maintenance safety backup, runs DBCC CHECKDB, rebuilds indexes with DBCC DBREINDEX at FillFactor 80, updates stats (sp_updatestats), and reports metrics.

2. System Requirements & Pre-Installation Checklist

Requirement	Specification
Operating System	Windows Server 2012 R2, 2016, 2019, 2022, 2025, or Windows 10/11 (64-bit)
Permissions	Local Administrator rights (required for Windows Task Scheduler and DPAPI vault)
Database Engine	Microsoft SQL Server 2012 through 2025 (Express, Standard, or Enterprise)
PowerShell Version	Windows PowerShell 5.1 (built into Windows) or PowerShell 7+
Network Connectivity	Outbound HTTPS (Port 443) to script.google.com and

	drive.google.com
Client Python Requirement	ABSOLUTELY NOT REQUIRED (Zero Runtime footprint)
Disk Space	50 MB for PowerShell agent; staging directory requires ~2.5x database size

3. Installation Track 1: Native Windows PowerShell Agent (Recommended)

This is the recommended, lightweight deployment track for servers. It uses out-of-the-box Windows OS tools with zero external binaries or compilers.

1. Step 1: Extract the provided client package zip file to a local staging folder, such as C:\Temp\Client_Package.
2. Step 2: Open an elevated PowerShell prompt: Click Windows Start -> search 'PowerShell' -> right-click 'Windows PowerShell' -> select 'Run as Administrator'.
3. Step 3: Navigate into the shell_client folder:
   cd C:\Temp\Client_Package\shell_client
4. Step 4: Execute the agent installer script:
   powershell.exe -ExecutionPolicy Bypass -File .\install_agent.ps1
5. Step 5: The installer validates the signed bundle.json, seals the machine authentication token into Windows DPAPI (LocalMachine scope), copies scripts into C:\Program Files\DatabaseBackupApp, and registers the 3 Windows Scheduled Tasks.
6. Step 6: You will see the green success banner confirming installation is complete.

```
cd C:\Temp\Client_Package\shell_client
powershell.exe -ExecutionPolicy Bypass -File .\install_agent.ps1
```

4. Installation Track 2: Graphical Setup Wizard (Setup_DatabaseBackup.exe)

If you prefer an interactive desktop wizard with graphical controls:

7. Step 1: In the extracted package root, locate Setup_DatabaseBackup.exe.
8. Step 2: Right-click Setup_DatabaseBackup.exe and select 'Run as administrator'.
9. Step 3: The setup wizard opens. Review the installation directory (defaults to C:\Program Files\DatabaseBackupApp).
10. Step 4: Confirm background schedule registration and desktop shortcuts.
11. Step 5: Click Install. The setup wizard copies runtime files, sets permissions, registers background tasks, and creates desktop/Start menu shortcuts.

5. Detailed Configuration Fields Reference: Required Values & Where to Find Them

Whether deploying via the Graphical Setup Wizard (Setup_DatabaseBackup.exe) or configuring the native PowerShell agent via C:\ProgramData\DatabaseBackupApp\config.json, each user-configurable field requires specific information from your local environment. Use the reference table and step-by-step guidance below to locate and enter every required detail:

Field / GUI Label	What to Fill (Format)	Where to Get It / How to Find It	Important Notes
SQL Server Instance (SQL_SERVER_INSTANCE)	Network hostname or SQL instance name. Examples: • localhost • .\SQLEXPRESS • SERVER01\SQL2019 • 192.168.1.50	1. SSMS Login Dialog: Look at 'Server name:' field. 2. Services.msc: Inspect 'SQL Server (...)' service. 3. CMD: Run 'hostname' command.	Default instance = 'localhost'. Named instances (e.g. Express) must include backslash (.\SQLEXPRESS).
Target Database(s) (DATABASE_NAME / TARGET_DATABASES)	Exact name(s) of the database to protect. Examples: • TheDatabase • GlassSheet, ERP_PROD	1. SSMS Object Explorer: Expand 'Databases' folder tree. 2. SQL Query: SELECT name FROM sys.databases WHERE database_id > 4; 3. ERP Software Vendor documentation.	Separate multiple databases with commas. Must match exact SQL database name.
SQL Authentication (SQL_USERNAME & SQL_PASSWORD)	SQL Login & Password. Examples: • Windows Auth: LEAVE BLANK • SQL Auth: 'sa' or 'backup_svc'	Provided by your internal Database Administrator (DBA). Must possess 'db_backupoperator' and 'db_datareader' (or sysadmin) rights.	Leaving blank uses Windows Integrated Authentication (-E), recommended for NT AUTHORITY\SYSTEM service accounts.
Backup Destination Folder (BACKUP_FOLDER / BACKUP_FOLDER_PATH)	Full local absolute drive path for staging compressed backups. Examples: • C:\SQLBackups • D:\DatabaseBackups	Open Windows File Explorer -> select a local drive with at least 2x the uncompressed database size free -> create a dedicated folder.	CRITICAL RESTRICTION: NEVER use OneDrive, Dropbox, or Google Drive folders. Cloud sync paths are blocked to prevent sync lockups.
Cloud Broker URL (BROKER_URL)	HTTPS Google Apps Script Web App URL. Example: https://script.google.com/macros/s/AKfycb.../exec	1. Automatically populated from the signed 'bundle.json' included in your package. 2. Or provided in onboarding documentation from your backup	Read-only in standard customer deployments; verified against Admin Ed25519 digital signature.

		service administrator.	
Customer Slug & Enrollment Token (CUSTOMER_SLUG / ENROLL_CODE)	Customer Slug: lowercase alphanumeric identifier (e.g. 'acme', 'fmi'). Enrollment Code: One-time machine token (e.g. 'ACME@b1357373.corp').	1. Automatically imported from package 'bundle.json'. 2. Or listed in Onboarding Credentials sheet Column A/C.	The installer exchanges this code once for a local DPAPI machine-bound token ('token.dpapi'). Never re-entered.
Automated Backup Schedule (SCHEDULE_DAYS & SCHEDULE_TIME)	Days: MON, WED, FRI or Daily Time: 24-hour HH:MM format. Examples: • Days: MON,TUE,WED,THU,FRI • Time: 02:00 (2:00 AM)	Determined by company IT policy. Schedule during off-peak maintenance hours when active user transactions are lowest.	Enforced by Windows Task Scheduler ('DatabaseBackup_Daily') with highest privileges (NT AUTHORITY\SYSTEM).
Local Retention & Cleanup (RETENTION_DAYS & CLEANUP_THRESHOLD_PERCENT)	Retention: Integer days (e.g. 7 or 30). Threshold: Integer percent (e.g. 85 or 90). Default: 30 days, 85% disk limit.	Internal company IT retention guidelines. Pre-configured with industry-standard defaults.	Module 2 monitors disk space hourly and purges staging backups older than RETENTION_DAYS, or triggers emergency purge if disk exceeds threshold.
Performance Maintenance (PERF_FILL_FACTOR & PERFORMANCE_SCHEDULE_DAY/TIME)	FillFactor: Integer 50-100 (Default: 80). Day: Sunday (or Daily/Weekly). Time: 03:30 (24h format).	Pre-configured to Microsoft recommended standard (80% FillFactor) for index rebuilds and statistics refresh.	Runs pre-maintenance safety backup, DBCC CHECKDB consistency checks, DBCC DBREINDEX, and sp_updatestats.

> **[PRO-TIP: LOCATING YOUR SQL SERVER INSTANCE]** HOW TO SAFELY LOCATE YOUR SQL INSTANCE NAME IN UNDER 30 SECONDS:
> • If SQL Server runs on this same server: Try 'localhost' or '.' first. If using SQL Express edition, use '.\SQLEXPRESS'.
> • To verify with 100% certainty: Press Win + R, type 'services.msc', and press Enter. Scroll down to 'SQL Server'. If the service name inside parentheses is 'MSSQLSERVER', your instance is 'localhost'. If it says anything else, such as 'SQL Server (PROD_INSTANCE)', your instance name is '.\PROD_INSTANCE'.

> **[CRITICAL STORAGE WARNING]** ANTI-DATA CORRUPTION WARNING — CLOUD SYNC DIRECTORIES ARE STRICTLY FORBIDDEN:
> Do NOT set the local Backup Destination Folder to any path containing 'OneDrive', 'Dropbox', 'Google Drive', or 'iCloud'. Active cloud synchronization engines attempt to lock files during differential write operations, leading to corrupted SQL database dumps and transmission aborts. The installation wizard will automatically reject cloud sync directories.

6. Manual Testing & Operational Verification

You can test each module immediately from an elevated PowerShell terminal:

1. Test Database Backup Module:

```
powershell.exe -ExecutionPolicy Bypass -File "C:\Program
Files\DatabaseBackupApp\backup_agent.ps1" -DatabaseName "TheDatabase"
```

2. Test Server Cleanup Scan Module:

```
powershell.exe -ExecutionPolicy Bypass -File "C:\Program
Files\DatabaseBackupApp\storage_monitor.ps1"
```

3. Test Performance Maintenance Module:

```
powershell.exe -ExecutionPolicy Bypass -File "C:\Program
Files\DatabaseBackupApp\performance_query.ps1" -DatabaseName "TheDatabase" -Mode Manual
```

4. Master Orchestrator (Runs All Modules):

```
powershell.exe -ExecutionPolicy Bypass -File "C:\Program
Files\DatabaseBackupApp\run_automation.ps1" -Task All -Mode Manual
```

7. Complete Step-by-Step Uninstallation Procedures

If you ever need to remove the software from the server, choose the uninstallation method that matches your deployment preference. Both methods perform complete, clean uninstallation with zero leftover background tasks.

Method 1: Native PowerShell Agent Uninstaller (Fastest & Cleanest)

12. Step 1: Open an elevated PowerShell prompt (Run as Administrator).
13. Step 2: Run the uninstaller script:
    powershell.exe -ExecutionPolicy Bypass -File "C:\Program Files\DatabaseBackupApp\uninstall_agent.ps1"
14. Step 3: Confirm with 'Y' when prompted.

15. Step 4: The script performs 6 clean phases:
    - Archives audit logs to Desktop (DatabaseBackup_Logs_Archive)
    - Terminates running backup and query processes
    - Unregisters all Windows Scheduled Tasks
    - Shreds DPAPI machine authentication tokens
    - Purges C:\Program Files\DatabaseBackupApp and C:\ProgramData\DatabaseBackupApp
    - Deletes all desktop and start menu shortcuts
16. Step 5: Displays green confirmation: 'UNINSTALLATION COMPLETE (100% CLEAN)'.

```
# Interactive Uninstallation (prompts before delete & archives logs)
powershell.exe -ExecutionPolicy Bypass -File "C:\Program
Files\DatabaseBackupApp\uninstall_agent.ps1"

# Silent Unattended Uninstallation (for RMM / Automated scripts)
powershell.exe -ExecutionPolicy Bypass -File "C:\Program
Files\DatabaseBackupApp\uninstall_agent.ps1" -Force
```

Method 2: Windows Settings / Control Panel / Start Menu

- Option A (Settings): Open Windows Settings -> Apps -> Installed apps -> locate 'Database Cloud Backup' -> click Uninstall.
- Option B (Start Menu): Click Windows Start -> locate 'Uninstall Database Cloud Backup' -> click to run.
- Option C (Batch File): Navigate to C:\Program Files\DatabaseBackupApp (or package root) -> double-click Uninstall.bat.
- Follow on-screen prompt -> Click Yes to confirm -> The uninstaller cleanly wipes tasks, shortcuts, registry entries, and program directories.

8. Log Files, Monitoring & Troubleshooting

Log File / Monitor	Description & Usage
C:\ProgramData\DatabaseBackupApp\logs\backup.log	Chronological log of database backups, compression ratios, and cloud uploads.
C:\ProgramData\DatabaseBackupApp\logs\storage_cleanup.log	Disk space scan results, free space percentages, and staging purge records.
C:\ProgramData\DatabaseBackupApp\logs\performance_query.log	Index fragmentation measurements (before/after), CHECKDB integrity output, and DBREINDEX duration.
Windows Task Scheduler (taskschd.msc)	Inspect Task Scheduler Library -> tasks prefixed with 'DatabaseBackup_' to review last run time and exit codes (0x0 = Success).